LSPC-LA: Local Structure Preserving Clustering With Learnable Anchors

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***Abstract*—K-means algorithm divides samples into c classes based on their structural characteristics. However, due to the non-convex nature of the clustering problem. algorithms are prone to converge to poor local minima. To address the aforementioned issues, we propose the Local Structure Preserving Clustering with Learnable Anchors (LSPC-LA) method. We assume that with a well-designed anchor selection strategy, samples near the same anchor tend to belong to the same cluster, which reveals high-confidence Must-Link local structural information for clustering. Based on this observation, we first construct the indicator matrix with prior information, which helps transform the sample clustering problem into an anchor clustering problem, thus reducing the solution space and minimizing the risk of poor local minima. Then, an anchor guiding matrix is constructed to enable anchors to capture the sample structure, improving clustering performance. Subsequently, an alternating iterative algorithm is proposed to optimize the LSPC-LA model. Finally, extensive experiments demonstrate the accuracy of the local structural information and the effectiveness of the LSPC-LA model.**

***Index Terms*—Clustering, learnable anchor, must-link information, label propagation.**

I. Introduction

THE K-means method [1], [2], [3] is a traditional clustering approach valued for its simplicity and scalability, leading to numerous optimized variants. However, due to its non-convex nature, the optimization problem of K-means is NP-hard [4], making it difficult to find the global optimum. The Lloyd heuristic algorithm [5] is widely used to address this problem. Nevertheless, the Lloyd heuristic algorithm has several drawbacks. For instance, its performance heavily depends on the initialization of centroids [6], and it is prone to getting trapped in suboptimal local minima [7].

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The code has been made publicly available on GitHub at: https://github.com/haonanxin/LSPC-LA_code.

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Given the limitations of the Lloyd method in clustering, researchers have begun exploring methods that utilize the local structure information of samples to improve clustering performance. These methods can be classified into three main categories: The first category is graph-based clustering methods [8], [9], [10], which leverage the structural properties of graphs to emphasize local similarity and connectivity between nodes, thereby capturing the inherent structure of the samples more effectively. The CSLP model [11] jointly leverages sample feature information and local structural information to learn an optimal graph, thereby achieving highly competitive clustering performance. The second category is density-based or similarity-based clustering methods [12], [13], [14], which focus on analyzing the density or similarity of local samples to identify clustering structures, especially in datasets with varying densities or non-uniform distributions. For example, by identifying both quality peaks and distance peaks, the TORC model [15] can effectively detect and eliminate erroneous merges, which is highly beneficial for improving clustering performance. The third category is multi-view and stream data clustering methods [16], [17], [18], which combine local information from multiple perspectives and apply local structure in dynamic and incremental learning scenarios, adapting to changing data environments, thereby enhancing the accuracy and robustness of clustering.

Although the aforementioned studies have achieved significant improvements in clustering performance, they have not effectively addressed the issue of K-means algorithm's susceptibility to poor local optima. To tackle this problem, we take another perspective by cleverly utilizing anchors to extract the inherent local structural information from the samples. This local structural information is used to narrow the solution space, thereby reducing the likelihood of getting trapped in poor local optima. Specifically, with a well-designed anchor selection strategy, samples that are closest to the same anchor tend to belong to the same cluster.[1] By leveraging this high-confidence Must-Link local structure information, we transform the original sample clustering problem into an anchor clustering problem, effectively shrinking the solution space and reducing the chances of falling into poor local minima.

However, existing anchor-based studies [19], [20], [21] often suffer from the issue that constructed anchors do not correspond to the clustering structure of the original samples. As illustrated

[1] We define the accuracy of local structural information and design an experiment to validate its effectiveness.

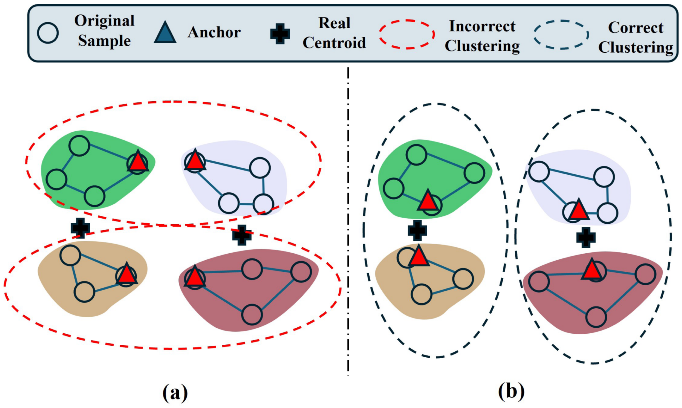


Fig. 1. The quality of anchors significantly impacts the clustering results.

in Fig. 1(a), the four anchors fail to correctly align with the original clustering structure, leading to incorrect clustering results. The propagation of local structure information further amplifies these errors, misclassifying all samples associated with these anchors and causing severe clustering deviations. To overcome this issue, this paper proposes an anchor self-learning approach, as shown in Fig. 1(b), where anchors are adjusted by learning the clustering structure of the samples, thereby enhancing the accuracy of anchor-based clustering.

Based on the above analysis, we propose the Local Structure Preserving Clustering with Learnable Anchors (LSPC-LA). As illustrated in Fig. 2, the indicator matrix with prior information is constructed based on anchors to capture the local structure information, which is reflected through Must-Link constraints. Then, a learnable anchor guiding matrix is developed to refine the anchor structure, aligning it with the clustering structure of the samples. Therefore, LSPC-LA not only narrows the solution space to reduce the likelihood of converging to poor local minima but also ensures the performance of anchor-based clustering. The main contributions of this paper are as follows:

- Unlike previous anchor-based clustering, where anchors are only used to represent surrounding samples to reduce computational complexity, our work effectively utilizes anchors to extract local structural information hidden within the samples. This transforms the clustering of the original samples into clustering on anchors, thereby reducing the solution space and decreasing the likelihood of convergence to poor local minima. Extensive experiments validate the accuracy of the proposed local structure information.
- Using the indicator matrix with prior information, a learnable anchor guiding matrix is proposed to refine the anchor structure, ensuring anchor alignment with the clustering structure of the samples and enhancing the performance of anchor-based clustering.
- A comprehensive optimization framework is designed to solve the optimization problem of LSPC-LA. Extensive experiments on real-world datasets demonstrate the advanced clustering performance.

Notations: In this paper, uppercase bold letters (e.g., $\boldsymbol{X}$) represent matrices, while lowercase letters (e.g., x) represent vectors or scalars. $\boldsymbol{X}_{i,:}$ denotes the i-th row of $\boldsymbol{X}$, $\boldsymbol{X}_{:,j}$ denotes the j-th column of $\boldsymbol{X}$, and $\boldsymbol{X}_{i,j}$ represents the (i, j)-th element of $\boldsymbol{X}$. For clarity, Table I presents the main notations throughout this work.

TABLE I
MAIN NOTATIONS

Notations	Descriptions
$d \in \mathbb{N}^+$	The feature dimension of samples
$n \in \mathbb{N}^+$	The number of samples
$c \in \mathbb{N}^+$	The number of classes
$m \in \mathbb{N}^+$	The number of anchors
$\mathbf{X} \in \mathbb{R}^{d \times n}$	The data matrix
$\mathbf{P} \in \mathbb{R}^{d \times m}$	The anchor matrix
$\mathbf{M} \in \mathbb{R}^{n \times m}$	The indicator matrix with prior information
$\mathbf{K} \in \mathbb{R}^{m \times m}$	The normalization matrix
$\mathbf{Q} \in \mathbb{R}^{n \times m}$	The anchor guiding matrix
$\mathbf{S} \in \mathbb{R}^{m \times c}$	The indicator matrix for anchors
$\mathbf{R} \in \mathbb{R}^{d \times c}$	The cluster center matrix of anchors

II. Related Work

A. Local Structure Learning

Local structure is essential in clustering, as it enables more precise identification of data patterns, improving accuracy and resolution. By minimizing global information interference, it aligns the clustering results with the true distribution of the samples. Effectively leveraging sample structures is crucial for enhancing clustering performance.

Many studies have extensively explored the local structure in clustering problems. For example, Zhang et al. [22] used local linear embedding in multi-view clustering to preserve the linear relationships between samples and their neighborhoods, ensuring local geometric invariance. They also employed laplacian eigenmaps to capture local consistency between tasks and views, enhancing clustering performance. Additionally, UMAP [23] optimizes local neighborhood structures to retain key features during clustering, thereby improving performance. Furthermore, diffusion maps [24], combined with spatial regularization and smooth stochastic processes, enhance the manifold structure capture in K-means, revealing the complex relationship between local and global structures in clustering.

However, despite existing studies focusing on enhancing the structural representation of data or establishing the connection between local and global structures, these methods still fail to effectively mitigate the issue of K-means algorithms getting stuck in poor local optima.

B. Anchor Theory

In recent years, anchor-based clustering algorithms have attracted significant research attention due to the important role that anchors play in improving clustering performance. Anchors serve as representatives of the data, simplifying the representation of the original samples [25], [26]. Specifically, the original dataset consists of a set of samples, while the anchors form a much smaller set that captures the essential structure of the data. To model the relationships between anchors and the original samples, anchor graphs are commonly employed. An anchor

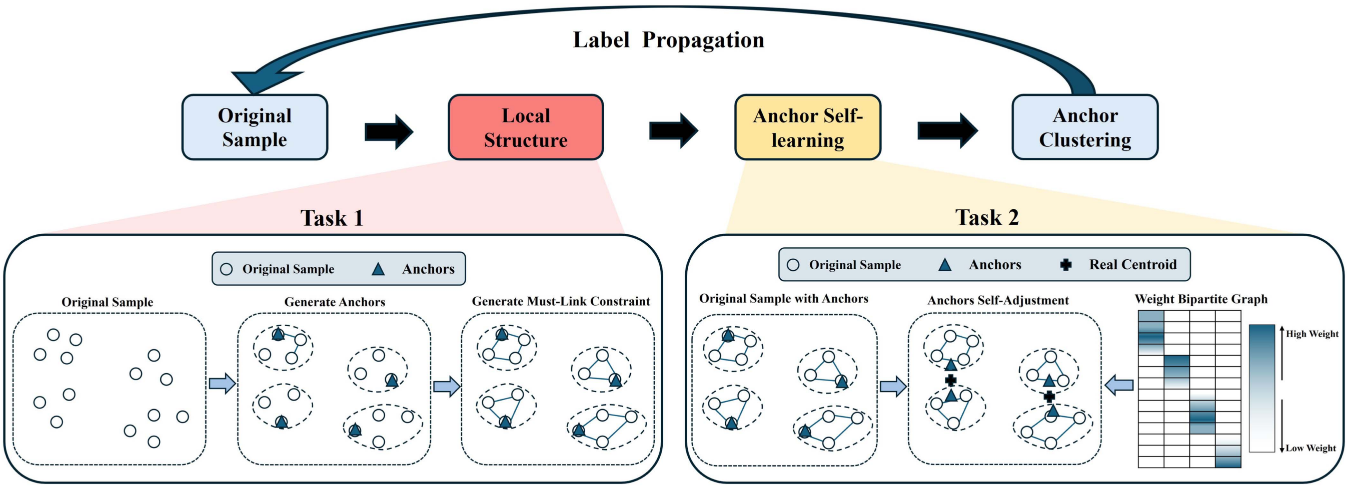


Fig. 2. The LSPC-LA consists of two key components: narrowing the solution space using local structure information, and learning the clustering structure of the samples. By narrowing the solution space, the probability of the algorithm getting trapped in suboptimal local minima is reduced. Learning the clustering structure of the samples ensures that the anchors maintain consistent clustering structures with the samples, thereby enhancing the accuracy of anchor-based clustering.

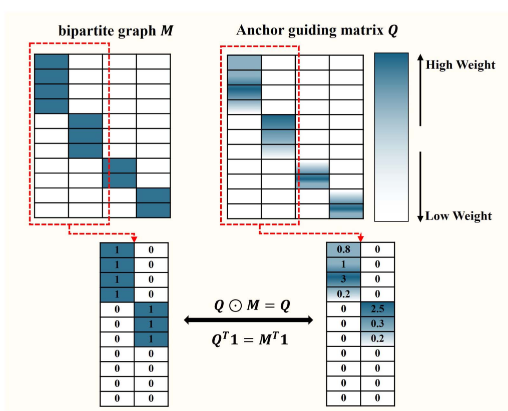
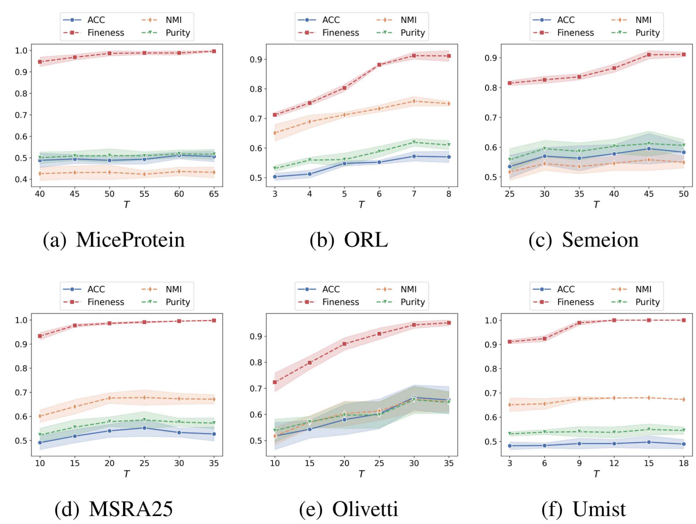


Fig. 3. The key properties of anchor guiding matrix $\boldsymbol{Q}$.



(a) MiceProtein (b) ORL (c) Semeion

(d) MSRA25 (e) Olivetti (f) Umist

Fig. 4. The quality of the must-link constraint information extracted by LSPC-LA and its clustering performance under different numbers of anchors.

graph [27] is a type of graph in which the nodes are divided into two distinct subsets, and edges only exist between nodes from different subsets, representing the connections between anchors and samples.

In anchor-based clustering, anchors simplify computational complexity [28], [29] by representing the core sample structure. This allows the construction of a mapping matrix between the data and anchors, approximating the spectral embedding and accelerating processing speed [30]. The second purpose is to enhance the representational capacity of the data. Specifically, anchors can integrate multi-view data by embedding information from different views into the anchor structure [31]. The third purpose is to mitigate the impact of noise on clustering performance. In multi-view clustering, Li et al. [32] enhanced robustness to data noise and inter-view differences by incorporating cross-view consistency constraints. On the other hand, Wang et al. [33] used a local-global fuzzy approach based on anchors to reduce noise interference in clustering.

Nevertheless, in existing anchor-based clustering methods, the construction of anchors is isolated from the clustering task. Once the anchors are constructed, they remain fixed and are not updated during the clustering process. As a result, the anchors cannot adjust their structure according to the clustering process, failing to maintain consistency with the clustering structure of the original samples.

C. Anchor Learning Methods

A growing line of work recognizes that the quality of anchor-based clustering hinges on how well anchors reflect the intrinsic data geometry, rather than on the efficiency of their selection. Heuristic anchors may be sufficient for roughly convex, uniformly sampled data, but they can drift from class modes when cluster densities are nonuniform or clusters are multimodal, undermining both representation and partition quality. This has motivated methods that learn anchors and their connectivity to better align with the true structure. In multi-view settings, some approaches jointly learn anchor graphs potentially with view-specific sizes while aligning view representations in a shared subspace [34], [35], [36], [37]; others adopt a consensus perspective, learning a near block-diagonal consensus anchor graph to improve stability and efficiency for multi-view subspace clustering [38]. For single-view large-scale clustering, recent work explicitly optimizes anchor graphs with objectives that maximize within-cluster compactness and enforce between-cluster separation, achieving a balance between speed and accuracy [21]. Deep variants couple autoencoder- or contrastive-based representation learning with anchor-graph optimization, using deep features to improve anchor coverage and separability while imposing global structural constraints via the anchor graph [39]. End-to-end formulations that co-adapt features and anchor relations help mitigate the mismatch caused by "feature-then-cluster" or "anchor-then-cluster" pipelines. Rather than relying on fixed heuristics, we constrain and regularize the anchor–sample solution space so that anchors can self-adjust toward class-consistent modes while preserving efficiency.

III. Methodology

A. The Proposed Model

1) Module 1. Local Structure Extraction: Assumption 1: With a well-designed anchor selection strategy, samples closest to the same anchor typically belong to the same cluster.[1]

Based on Assumption 1, we reformulate the sample clustering as an anchor clustering, leveraging the relationships between samples and anchors that represent the local structure of the samples. This allows us to construct an indicator matrix with prior information, which encodes Must-Link constraints, capturing the local consistency between similar samples. The Must-Link information ensures that samples within the same cluster are connected and share similar characteristics. Using label propagation, we propagate labels from anchors to the original samples based on these constraints. By learning the anchor label matrix, we ultimately determine the clustering labels of the original samples.

Given dataset $\boldsymbol{X} = [x_1, x_2, \ldots, x_n] \in \mathbb{R}^{d\times n}$, where x_i represents the i-th sample, the goal of K-means is to partition $\boldsymbol{X}$ into c groups, and the objective is expressed as

$$\min_{\boldsymbol{F}\in Ind, \boldsymbol{H}} \left\|\boldsymbol{X} - \boldsymbol{H}\boldsymbol{F}^T\right\|_F^2, \tag{1}$$

where $\boldsymbol{H} \in \mathbb{R}^{d\times c}$ denotes the cluster center matrix in sample space, and $\boldsymbol{F} \in \mathbb{R}^{n\times c}$ represents the indicator matrix of samples. However, the aforementioned methods overlook the exploitation of local structural information in clustering, which inherently limits their clustering performance.

In order to solve the above problem, rather than conducting clustering directly on the data matrix $\boldsymbol{X}$, our approach performs clustering at the anchor level. By employing the K-means++ algorithm, m subclusters can be identified, and accordingly, an indicator matrix $\boldsymbol{M} \in \mathbb{R}^{n\times m}$ mapping samples to anchors, which preserves rich local structural relationships, can be constructed. Furthermore, through the adaptive anchor learning strategy proposed in Module 2, the anchor matrix $\boldsymbol{P} \in \mathbb{R}^{d\times m}$ can be effectively learned. Consequently, the clustering model is formulated by applying K-means directly to the learnable anchors $\boldsymbol{P}$ as follows

$$\min_{\boldsymbol{S}\in Ind, \boldsymbol{R}} \left\|\boldsymbol{P} - \boldsymbol{R}\boldsymbol{S}^T\right\|_F^2. \tag{2}$$

After obtaining the learned anchor label matrix $\boldsymbol{S} \in \mathbb{R}^{m\times c}$, the clustering labels of the original samples can be obtained through label propagation as $\boldsymbol{F} = \boldsymbol{M}\boldsymbol{S}$.

2) Module 2. Adaptive Anchor Learning: Based on Section II-B, we expect the anchors to learn the sample structure during the clustering process and adjust accordingly to maintain a clustering structure consistent with the original samples.

Based on this idea, we propose the anchor guiding matrix $\boldsymbol{Q}$, which satisfies the constraint $\boldsymbol{Q^T 1} = \boldsymbol{M^T 1}$. As an adaptive alternative to $\boldsymbol{M}$, $\boldsymbol{Q}$ allows non-uniform sample contributions to anchor positions, enabling anchors to self-adjust based on sample structure and improving cluster separability. In addition, to ensure the position weight of an anchor should be determined by the contributions of the samples it represents, we impose a Hadamard product constraint $\boldsymbol{Q} \odot \boldsymbol{M} = \boldsymbol{Q}$, using the previously constructed matrix $\boldsymbol{M}$ as prior information. This maintains structural consistency between $\boldsymbol{Q}$ and $\boldsymbol{M}$. A conceptually natural model can be defined in mathematical form as follows

$$\begin{aligned}&\min_{\boldsymbol{P},\boldsymbol{Q}} \left\|\boldsymbol{X} - \boldsymbol{P}\boldsymbol{Q}^T\right\|_F^2 + \lambda \left\|\boldsymbol{Q}\right\|_F^2,\\ &\quad \text{s.t.} \quad \boldsymbol{Q} \odot \boldsymbol{M} = \boldsymbol{Q},\ \boldsymbol{Q}^T\boldsymbol{1} = \boldsymbol{M}^T\boldsymbol{1},\ \boldsymbol{Q} \geq 0,\end{aligned} \tag{3}$$

where $\|\boldsymbol{Q}\|_F^2$ serves as a regularization term.

According to the Lloyd algorithm's approach for cluster center calculation, we have

$$\boldsymbol{P} = \boldsymbol{X}\boldsymbol{Q}(\boldsymbol{Q}^T\boldsymbol{Q})^{-1}, \tag{4}$$

where $\boldsymbol{XQ}$ represents the accumulation of samples assigned to their clusters. However, the diagonal entries of $\boldsymbol{Q^T Q}$ no longer correspond to the exact number of samples assigned to each subcluster. In this case, due to the fuzzy nature of the matrix $\boldsymbol{Q}$, which differs from the binary nature of the indicator matrix $\boldsymbol{M}$ with values in $\{0, 1\}$, the data scale is magnified when updating the anchors $\boldsymbol{P}$. We provide a proof of this result in Lemma 1.

To address the aforementioned issue, we introduce a normalization matrix $\boldsymbol{K} \in \mathbb{R}^{m\times m}$, which is mathematically defined as $\boldsymbol{K} = (diag(\boldsymbol{M}^T\boldsymbol{1}))^{-1}$. In this way, $\boldsymbol{XQK}^T$ represents the centroid matrix constructed using the indicator matrix $\boldsymbol{M}$. By enforcing its proximity to the learnable anchor matrix $\boldsymbol{P}$, the adaptive anchor learning module is formulated as follows

$$\begin{aligned}&\min_{\boldsymbol{P},\boldsymbol{Q}} \left\|\boldsymbol{P} - \boldsymbol{X}\boldsymbol{Q}\boldsymbol{K}^T\right\|_F^2 + \lambda \left\|\boldsymbol{Q}\right\|_F^2,\\ &\quad \text{s.t.} \quad \boldsymbol{Q} \odot \boldsymbol{M} = \boldsymbol{Q},\ \boldsymbol{Q}^T\boldsymbol{1} = \boldsymbol{M}^T\boldsymbol{1},\ \boldsymbol{Q} \geq 0,\end{aligned} \tag{5}$$

3) The LSPC-LA Model: The core idea of the model is to leverage the matrix $\boldsymbol{M}$ as prior information to construct an anchor guiding matrix $\boldsymbol{Q} \in \mathbb{R}^{n\times m}$, enabling the anchors to capture the local structural information of the original samples. Furthermore, we aim for the anchors to be learned adaptively in a manner that mutually promotes clustering, all within a unified optimization framework. Accordingly, the Local Structure Preserving Clustering with Learnable Anchors (LSPC-LA) model is defined as follows

$$\min_{\boldsymbol{P},\boldsymbol{R},\boldsymbol{S},\boldsymbol{Q}} \left\|\boldsymbol{P}-\boldsymbol{X}\boldsymbol{Q}\boldsymbol{K}^T\right\|_F^2+\mu\left\|\boldsymbol{P}-\boldsymbol{R}\boldsymbol{S}^T\right\|_F^2+\lambda\left\|\boldsymbol{Q}\right\|_F^2,$$
$$\text{s.t. } \boldsymbol{Q}\odot\boldsymbol{M}=\boldsymbol{Q},\ \boldsymbol{Q}^T\mathbf{1}=\boldsymbol{M}^T\mathbf{1},\ \boldsymbol{Q}\geq 0,$$
$$\boldsymbol{S}\in Ind,\ \boldsymbol{K}=(diag(\boldsymbol{M}^T\mathbf{1}))^{-1}. \tag{6}$$

Lemma 1: When $\boldsymbol{Q}$ is a matrix with $\boldsymbol{Q}\odot\boldsymbol{M}=\boldsymbol{Q}$ and $\boldsymbol{Q}^T\mathbf{1}=\boldsymbol{M}^T\mathbf{1}$ constraint rather than a $\{0,1\}$ indicator matrix, replacing $\|\boldsymbol{X}-\boldsymbol{P}\boldsymbol{Q}^T\|_F^2$ with $\|\boldsymbol{P}-\boldsymbol{X}\boldsymbol{Q}\boldsymbol{K}^T\|_F^2$ in the Problem (6) makes the data scale remain unchanged during optimization.

Proof: If the objective function is expressed in the form of (3), our primary focus is on its first term. Therefore, (3) is transformed into

$$\min_{\boldsymbol{P},\boldsymbol{Q}} \left\|\boldsymbol{X}-\boldsymbol{P}\boldsymbol{Q}^T\right\|_F^2,$$
$$\text{s.t. } \boldsymbol{Q}\odot\boldsymbol{M}=\boldsymbol{Q},\ \boldsymbol{Q}^T\mathbf{1}=\boldsymbol{M}^T\mathbf{1},\ \boldsymbol{Q}\geq 0. \tag{7}$$

When updating $\boldsymbol{P}$, we take its derivative, set it to zero and get

$$p_i=\boldsymbol{X}q_i(q_i^Tq_i)^{-1}, \tag{8}$$

where $\boldsymbol{Q}$ is essentially the fuzzy counterpart of $\boldsymbol{M}$. When $\boldsymbol{Q}$ is replaced by $\boldsymbol{M}$, (8) becomes consistent with the centroid calculation formula of Lloyd's algorithm for the K-means problem

$$p_i=\boldsymbol{X}m_i\left(m_i^Tm_i\right)^{-1}. \tag{9}$$

This implies that the samples within each cluster are averaged with weighted contributions, and normalized by $m_i^Tm_i$, where $m_i^Tm_i$ represents the number of samples in the i-th cluster. However, when $\boldsymbol{Q}$ is fuzzy, it follows that $q_i^Tq_i\geq m_i^Tm_i$. In this case, while the numerator still represents the weighted average of samples within the cluster, it is no longer normalized by the number of samples in the i-th subcluster. As a result, the scaling of the denominator is enlarged, leading to a reduction in the scale of $\boldsymbol{P}$ after the calculation.

When we express the objective in the form of (6), the primary focus remains on the first term. Therefore, (6) is transformed into

$$\min_{\boldsymbol{P},\boldsymbol{Q}} \left\|\boldsymbol{P}-\boldsymbol{X}\boldsymbol{Q}\boldsymbol{K}^T\right\|_F^2,$$
$$\text{s.t. } \boldsymbol{Q}\odot\boldsymbol{M}=\boldsymbol{Q},\ \boldsymbol{Q}^T\mathbf{1}=\boldsymbol{M}^T\mathbf{1},$$
$$\boldsymbol{Q}\geq 0,\ \boldsymbol{K}=(diag(\boldsymbol{M}^T\mathbf{1}))^{-1}. \tag{10}$$

When updating $\boldsymbol{P}$, we take its derivative, set it to zero and get

$$p_i=\boldsymbol{X}q_i(m_i^T\mathbf{1})^{-1}. \tag{11}$$

Obviously, the denominator still represents the number of samples in the i-th cluster, so it retains the original data scale, and $\boldsymbol{P}$ also maintains the original data scale. □

B. Analysis of LSPC-LA

Anchor Learnable: The learning of anchors is primarily reflected in the anchor guiding matrix $\boldsymbol{Q}$. Fig. 3 illustrates its key properties. Each column of $\boldsymbol{Q}$ has a fixed sum, representing the number of samples covered by each anchor, ensured by the constraint $\boldsymbol{Q}^T\mathbf{1}=\boldsymbol{M}^T\mathbf{1}$. The contribution weights of samples to their anchors vary. This adaptive weight updating enables the anchor structure to be learnable during clustering.

Anchor Stability: As shown in Fig. 3, the Hadamard product constraint ensures that $\boldsymbol{Q}$ maintains the same structure as $\boldsymbol{M}$, meaning positions where $\boldsymbol{M}$ is zero are also zero in $\boldsymbol{Q}$. This constraint prevents samples not represented by an anchor from influencing it, thus restricting anchor positions to their respective covered samples and enhancing anchor learning stability.

IV. Optimization and Theoretical Analysis

A. Optimization

For Problem (6), there are four variables, $\boldsymbol{P}$, $\boldsymbol{R}$, $\boldsymbol{S}$, and $\boldsymbol{Q}$, that need to be solved. To address this, we employ an alternating iterative optimization approach, which decomposes the problem into the following subproblems.

Step 1. Update $\boldsymbol{P}$ with fixed $\boldsymbol{Q}$, $\boldsymbol{S}$ and $\boldsymbol{R}$: The problem (6) can be derived to the following problem

$$\min_{\boldsymbol{P}} \left\|\boldsymbol{P}-\boldsymbol{X}\boldsymbol{Q}\boldsymbol{K}^T\right\|_F^2+\mu\left\|\boldsymbol{P}-\boldsymbol{R}\boldsymbol{S}^T\right\|_F^2. \tag{12}$$

Taking the derivative w.r.t. $\boldsymbol{P}$ and setting it to zero, we have

$$\boldsymbol{P}=\boldsymbol{X}\boldsymbol{Q}\boldsymbol{K}((1+\mu)\boldsymbol{I}-\mu\boldsymbol{S}(\boldsymbol{S}^T\boldsymbol{S})^{-1}\boldsymbol{S}^T)^{-1}. \tag{13}$$

Step 2. Update $\boldsymbol{Q}$ with fixed $\boldsymbol{P}$, $\boldsymbol{S}$ and $\boldsymbol{R}$: The problem (6) can be derived to the following problem

$$\min_{\boldsymbol{Q}} Tr(\boldsymbol{Q}^T\boldsymbol{B}\boldsymbol{Q}\boldsymbol{G})-2Tr(\boldsymbol{P}^T\boldsymbol{X}\boldsymbol{Q}\boldsymbol{K})+\lambda Tr(\boldsymbol{Q}^T\boldsymbol{Q}),$$
$$\text{s.t. } \boldsymbol{Q}\odot\boldsymbol{M}=\boldsymbol{Q},\ \boldsymbol{Q}^T\mathbf{1}=\boldsymbol{M}^T\mathbf{1},\ \boldsymbol{Q}\geq 0,$$
$$\boldsymbol{K}=(diag(\boldsymbol{M}^T\mathbf{1}))^{-1}, \tag{14}$$

where $\boldsymbol{B}=\boldsymbol{X}^T\boldsymbol{X}$ and $\boldsymbol{G}=\boldsymbol{K}\boldsymbol{K}^T$. For the first term in (14), it can be easily obtained that

$$\boldsymbol{Q}^T\boldsymbol{B}\boldsymbol{Q}\boldsymbol{G}=\begin{bmatrix} q_1^T\boldsymbol{B}q_1 & q_1^T\boldsymbol{B}q_2 & \dots & q_1^T\boldsymbol{B}q_m \\ q_2^T\boldsymbol{B}q_1 & q_2^T\boldsymbol{B}q_2 & \dots & q_2^T\boldsymbol{B}q_m \\ \vdots & \vdots & \ddots & \vdots \\ q_m^T\boldsymbol{B}q_1 & q_m^T\boldsymbol{B}q_2 & \dots & q_m^T\boldsymbol{B}q_m \end{bmatrix} \begin{bmatrix} \boldsymbol{G}_{1,1} & & & \\ & \boldsymbol{G}_{2,2} & & \\ & & \ddots & \\ & & & \boldsymbol{G}_{m,m} \end{bmatrix}, \tag{15}$$

$$Tr(\boldsymbol{Q}^T\boldsymbol{B}\boldsymbol{Q}\boldsymbol{G})=\sum_{i=1}^{m}\boldsymbol{G}_{i,i}q_i^T\boldsymbol{B}q_i. \tag{16}$$

For the second and third items of (14), we can obtain

$$2Tr(\boldsymbol{K}\boldsymbol{P}^T\boldsymbol{X}\boldsymbol{Q})=Tr(\boldsymbol{N}\boldsymbol{Q})=\sum_{i=1}^{m}\boldsymbol{N}_{i,:}q_i \tag{17}$$

Algorithm 1: The procedure to solve problem (6).

Input: Sample matrix $\boldsymbol{X}$, the indicator matrix $\boldsymbol{M}$, anchor matrix $\boldsymbol{P}$, matrix $\boldsymbol{K}$, the number of clusters c and the parameters λ and μ.
Output: The label matrix $\boldsymbol{F}$
1: Initialize the anchor cluster indicator matrix $\boldsymbol{S}$. Initialize the anchor guiding matrix $\boldsymbol{Q}$ under the constraint of $\boldsymbol{Q} \odot \boldsymbol{M} = \boldsymbol{Q}$. Get $\boldsymbol{B}$ by $\boldsymbol{B} = \boldsymbol{X^T X}$ and $\boldsymbol{G}$ by $\boldsymbol{G} = \boldsymbol{K K}^T$.
2: **while** not converged **do**
3: Update $\boldsymbol{P}$ by (13);
4: Update $\overline{q}_i$ by (20). Update $\boldsymbol{Q}$ by (21) and (22);
5: Update $\boldsymbol{S}$ through Algorithm 2;
6: **end while**
Obtain the labels $\boldsymbol{F}$ by $\boldsymbol{F} = \boldsymbol{MS}$
7: **return** $\boldsymbol{F}$

and

$$\lambda Tr(\boldsymbol{Q^T Q}) = \lambda \sum_{i=1}^{m} q_i^T q_i. \tag{18}$$

where $\boldsymbol{N} = 2\boldsymbol{KP^T X}$. Since each column of $\boldsymbol{Q}$ is independent, we update $\boldsymbol{Q}$ column by column. Based on the above analysis, for each column $\boldsymbol{q}_i$, we have

$$\begin{aligned} &\boldsymbol{G}_{i,i} q_i^T \boldsymbol{B} q_i - \boldsymbol{N}_{i,:} q_i + \lambda q_i^T q_i \\ &= q_i^T (\boldsymbol{G}_{i,i}\boldsymbol{B} + \lambda \boldsymbol{I}) q_i - \boldsymbol{N}_{i,:} q_i \\ &= q_i^T \boldsymbol{A^{(i)}} q_i - \boldsymbol{N}_{i,:} q_i, \end{aligned} \tag{19}$$

where $\boldsymbol{A^{(i)}} = \boldsymbol{G}_{i,i}\boldsymbol{B} + \lambda \boldsymbol{I}$.

For the constraint $\boldsymbol{Q} \odot \boldsymbol{M} = \boldsymbol{Q}$, the implication is that the update of $\boldsymbol{Q}$ only involves the entries corresponding to the nonzero positions in $\boldsymbol{M}$, which ensures that all elements of $\boldsymbol{Q}$ at the zero positions of $\boldsymbol{M}$ remain zero. Therefore, for each column q_i, we apply a mapping to obtain $\overline{q_i}$, calculated as follows

$$\overline{q_i} \leftarrow q_{i\psi(m_i)}, \tag{20}$$

where $\psi(m_i)$ denotes the indices corresponding to the nonzero entries of m_i. For example, if $a = [1, 0, 1, 1, 0]^T$, then $\psi(a) = [1, 3, 4]^T$. Therefore, problem (14) can be transformed into updating each column $\overline{q_i}$

$$\begin{aligned} &\min_{\overline{q_i}} \overline{q_i}^T \overline{\boldsymbol{A^{(i)}}}\, \overline{q_i} - \overline{\boldsymbol{N}}_{i,:} \overline{q_i}, \\ &\text{s.t.} \quad \overline{q_i}^T \boldsymbol{1} = m_i^T \boldsymbol{1},\ \overline{q_i} \geq 0, \end{aligned} \tag{21}$$

where $\overline{\boldsymbol{A^{(i)}}}$ and $\overline{\boldsymbol{N}}$ are also obtained by mapping $\boldsymbol{A^{(i)}}$ and $\boldsymbol{N}$ in the same way as described above. Obviously, the problem (21) can be solved by the standard QP solver [40]. So we can use the above method to update each $\overline{q_i}$ until convergence. Then, the final $\boldsymbol{Q}$ could be updated by

$$q_{i\psi(m_i)} \leftarrow \overline{q_i}. \tag{22}$$

Step 3. Update $\boldsymbol{R}$ and $\boldsymbol{S}$ with fixed $\boldsymbol{P}$ and $\boldsymbol{Q}$: The problem (6) can be derived to the following problem

$$\begin{aligned} &\min_{\boldsymbol{R,S}} \left\| \boldsymbol{P} - \boldsymbol{RS}^T \right\|_F^2, \\ &\text{s.t.} \quad \boldsymbol{S} \in Ind. \end{aligned} \tag{23}$$

In Problem (23), we focus on $\boldsymbol{S}$ rather than $\boldsymbol{R}$. The update formula for $\boldsymbol{R}$ follows the same form as in (4). Therefore, Problem (23) can be reformulated as:

$$\min_{\boldsymbol{S}\in Ind} Tr(\boldsymbol{PP}^T) - Tr(\boldsymbol{S}^T \boldsymbol{P}^T \boldsymbol{PS}(\boldsymbol{S}^T \boldsymbol{S})^{-1}). \tag{24}$$

Here, $Tr(\boldsymbol{PP}^T)$ is a constant, thus problem (24) can be transformed into

$$\max_{\boldsymbol{S}\in Ind} Tr(\boldsymbol{S}^T \boldsymbol{P}^T \boldsymbol{PS}(\boldsymbol{S}^T \boldsymbol{S})^{-1}). \tag{25}$$

Obviously, the problem (25) can be solved by the standard Coordinate Descent algorithm [41]. Let $\boldsymbol{S} = [s_1, s_2, \ldots, s_c]$, where s_l represents the l-th column of $\boldsymbol{S}$. Thus, we have

$$\boldsymbol{S}^T \boldsymbol{P}^T \boldsymbol{PS} = \begin{bmatrix} s_1^T \boldsymbol{P}^T \boldsymbol{P} s_1 & \cdots & s_1^T \boldsymbol{P}^T \boldsymbol{P} s_c \\ \vdots & \ddots & \vdots \\ s_c^T \boldsymbol{P}^T \boldsymbol{P} s_1 & \cdots & s_c^T \boldsymbol{P}^T \boldsymbol{P} s_c \end{bmatrix}, \tag{26}$$

and

$$\boldsymbol{S}^T \boldsymbol{S} = \begin{bmatrix} s_1^T s_1 & 0 & \cdots & 0 \\ 0 & s_2^T s_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & s_c^T s_c \end{bmatrix}. \tag{27}$$

Substituting (26) and (27) into (25), and noting that $\boldsymbol{S}^T \boldsymbol{S}$ is a diagonal matrix, (25) can be written as

$$\max_{\boldsymbol{S}\in Ind} \sum_{l=1}^{c} \frac{s_l^T \boldsymbol{P}^T \boldsymbol{P} s_l}{s_l^T s_l}. \tag{28}$$

For (28), since each row is independent, the coordinate descent algorithm [41] can be used to update the rows of the anchor cluster indicator matrix $\boldsymbol{S}$ sequentially. To update the i-th row of $\boldsymbol{S}$, there are c possible configurations for this row, where the entry 1 can be in any of the c positions, with all other entries being 0. Let $v \in \{1, 2, \ldots, c\}$ be the position of the 1in the i-th row, indicating the v-th configuration of $\boldsymbol{S}$ denoted as $\boldsymbol{S}^{(v)}$. Thus, problem (25) can be transformed into problem (29)

$$\begin{aligned} &\max_{\boldsymbol{S}^{(v)}\in Ind} \sum_{l=1}^{c} \frac{s_l^{(v)T} \boldsymbol{P}^T \boldsymbol{P} s_l^{(v)}}{s_l^{(v)T} s_l^{(v)}} \\ &\text{s.t.} \quad \boldsymbol{S}^{(v)} \in \{\boldsymbol{S}^{(1)}, \boldsymbol{S}^{(2)}, \ldots, \boldsymbol{S}^{(c)}\}, \end{aligned} \tag{29}$$

where $s_l^{(v)}$ is the l-th column of $\boldsymbol{S}^{(v)}$. During the iteration process, we can find the optimal solution of problem (29) by evaluating all possible configurations of $\boldsymbol{S}^{(v)}$. However, evaluating all configurations of $\boldsymbol{S}^{(v)}$ to solve problem (29) iintroduces a lot of redundancy, leading to a very high time complexity.

Revisiting problem (29), it is evident that for $\boldsymbol{S}^{(v)}$ and $\boldsymbol{S}^{(l)}$, the i-th row differs only at the v-th and l-th positions, with all other positions being identical and set to 0. Based on this observation, we introduce $\boldsymbol{S}^{(0)}$, which is identical to $\boldsymbol{S}^{(v)}$ except that all elements in the i-th row are 0. Thus, problem (29) can be transformed into

$$\max_{\boldsymbol{S}^{(v)}\in Ind} \frac{s_v^{(v)T}\boldsymbol{P}^T\boldsymbol{P}s_v^{(v)}}{s_v^{(v)T}s_v^{(v)}} - \frac{s_v^{(0)T}\boldsymbol{P}^T\boldsymbol{P}s_v^{(0)}}{s_v^{(0)T}s_v^{(0)}}$$
$$\text{s.t.}\ \ \boldsymbol{S}^{(v)} \in \{\boldsymbol{S}^{(1)}, \boldsymbol{S}^{(2)}, \ldots, \boldsymbol{S}^{(c)}\}. \tag{30}$$

For (30), we observe that all $\boldsymbol{S}^{(v)}$ differ only in the elements of the i-th row. To save space complexity, we can represent $\boldsymbol{S}^{(v)}$ by replacing the v-th column of the current anchor indicator matrix $\boldsymbol{S}$. Additionally, some computational techniques can be applied to further reduce time complexity.

Assuming that the current anchor cluster indicator matrix $\boldsymbol{S}$ is being updated for the i-th row, and the current position of 1 is k. There are two cases to consider: whether k is equal to v or not. We first introduce an intermediate variable $\mathit{\Gamma}$, which is a column vector with 1 at the i-th position and 0 at all other positions.

- $v = k$: This indicates that both s_v and $s_v^{(v)}$ have 1 at the i-th position, thus $s_v = s_v^{(v)}$, which means $s_v = s_v^{(0)} + \mathit{\Gamma}$. Therefore, we can derive the four components of problem (30) by

$$\boldsymbol{P}s_v^{(0)} = \boldsymbol{P}(s_v - \mathit{\Gamma}) = \boldsymbol{P}s_v - \boldsymbol{P}_{:,i}, \tag{31}$$
$$s_v^{(0)T}s_v^{(0)} = (s_v - \mathit{\Gamma})^T(s_v - \mathit{\Gamma}) = s_v^T s_v - 1, \tag{32}$$
$$\boldsymbol{P}s_v^{(v)} = \boldsymbol{P}s_v, \tag{33}$$
$$s_v^{(v)T}s_v^{(v)} = s_v^T s_v. \tag{34}$$

In this case, problem (30) can be transformed into

$$\mathcal{J}(\boldsymbol{S}^{(v)}) = \frac{s_v^T\boldsymbol{P}^T\boldsymbol{P}s_v}{s_v^T s_v} - \frac{s_v^T\boldsymbol{P}^T\boldsymbol{P}s_v - 2\boldsymbol{P}_{:,i}^T\boldsymbol{P}s_v + (\boldsymbol{P}^T\boldsymbol{P})_{i,i}}{s_v^T s_v - 1}, \tag{35}$$

where $\boldsymbol{P}_{:,i}$ denotes the i-th column of $\boldsymbol{P}$ and $(\boldsymbol{P}^T\boldsymbol{P})_{i,i}$ denotes the (i,i)-th element of $(\boldsymbol{P}^T\boldsymbol{P})$.

- $v \neq k$: This indicates that both $s_v^{(v)}$ and $s_v^{(0)}$ have 0 at the i-th position, thus $s_v = s_v^{(0)}$, which means $s_v = s_v^{(v)} - \mathit{\Gamma}$. Therefore, we can derive the four components of problem (30) by

$$\boldsymbol{P}s_v^{(v)} = \boldsymbol{P}(s_v + \mathit{\Gamma}) = \boldsymbol{P}s_v + \boldsymbol{P}_{:,i}, \tag{36}$$
$$s_v^{(v)T}s_v^{(v)} = (s_v + \mathit{\Gamma})^T(s_v + \mathit{\Gamma}) = s_v^T s_v + 1, \tag{37}$$
$$\boldsymbol{P}s_v^{(0)} = \boldsymbol{P}s_v, \tag{38}$$
$$s_v^{(0)T}s_v^{(0)} = s_v^T s_v. \tag{39}$$

In this case, problem (30) can be transformed into

$$\mathcal{J}(\boldsymbol{S}^{(v)}) = \frac{s_v^T\boldsymbol{P}^T\boldsymbol{P}s_v + 2\boldsymbol{P}_{:,i}^T\boldsymbol{P}s_v + (\boldsymbol{P}^T\boldsymbol{P})_{i,i}}{s_v^T s_v + 1} - \frac{s_v^T\boldsymbol{P}^T\boldsymbol{P}s_v}{s_v^T s_v}. \tag{40}$$

Algorithm 2: The procedure to solve problem (28).

Input: The anchor matrix $\mathbf{P}\in\mathbb{R}^{d\times m}$ and the initial anchor indicator matrix $\mathbf{S}\in Ind$
Input: The anchor indicator matrix $\boldsymbol{S}\in Ind$
for $k = 1, 2, \ldots, c$ **do**
  Compute and store $\boldsymbol{P}s_k$, $s_k^T s_k$, $s_k^T\boldsymbol{P}^T\boldsymbol{P}s_k$.
end for
while not converge **do**
  for $i = 1, 2, \ldots, m$ **do**
    Let k be the current position of 1in the i-th row.
    for $v = 1, 2, \ldots, c$ **do**
      Calculate $\mathcal{J}(\boldsymbol{S}^{(v)})$ by (35) or (40)
    end for
    $v^* = \arg\max_{\boldsymbol{S}^{(v)}\in Ind}\mathcal{J}(\boldsymbol{S}^{(v)})$
    Update the i-th row of $\boldsymbol{S}$ by (42)
    if $k \neq v^*$ **then**
      Update $\boldsymbol{P}s_k$, $s_k^T s_k$, $\boldsymbol{P}s_{v^*}$ and $s_{v^*}^T s_{v^*}$ by (43) to (46)
      Update $s_k^T\boldsymbol{P}^T\boldsymbol{P}s_k$ and $s_{v^*}^T\boldsymbol{P}^T\boldsymbol{P}s_{v^*}$ by (47) and (48)
    end if
  end for
end while
return $\boldsymbol{S}$

Therefore, when updating the i-th row of $\boldsymbol{S}$, assume

$$v^* = \arg\max_{\boldsymbol{S}^{(v)}\in Ind}\mathcal{J}(\boldsymbol{S}^{(v)})$$
$$\text{s.t.}\ \ \boldsymbol{S}^{(v)} \in \{\boldsymbol{S}^{(1)}, \boldsymbol{S}^{(2)}, \ldots, \boldsymbol{S}^{(c)}\}, \tag{41}$$

we have

$$s_{ik} = \begin{cases} 1, & k = v^*; \\ 0, & otherwise. \end{cases}. \tag{42}$$

Before the c iterations of updating the i-th row of $\boldsymbol{S}$, the position of 1in the i-th row of $\boldsymbol{S}$ is k, then if $k = v^*$, it indicates that k is already the optimal position and no update is needed. If $k \neq v^*$, it means v^* is the optimal position and an update is required, thus

$$s_k^T s_k \Leftarrow s_k^T s_k - 1, \tag{43}$$
$$s_{v^*}^T s_{v^*} \Leftarrow s_{v^*}^T s_{v^*} + 1, \tag{44}$$
$$\boldsymbol{P}s_k \Leftarrow \boldsymbol{P}s_k - \boldsymbol{P}_{:,i}, \tag{45}$$
$$\boldsymbol{P}s_{v^*} \Leftarrow \boldsymbol{P}s_{v^*} + \boldsymbol{P}_{:,i}, \tag{46}$$
$$s_k^T\boldsymbol{P}^T\boldsymbol{P}s_k \Leftarrow s_k^T\boldsymbol{P}^T\boldsymbol{P}s_k - 2\boldsymbol{P}_{:,i}^T\boldsymbol{P}s_k + (\boldsymbol{P}^T\boldsymbol{P})_{i,i}, \tag{47}$$
$$s_{v^*}^T\boldsymbol{P}^T\boldsymbol{P}s_{v^*} \Leftarrow s_{v^*}^T\boldsymbol{P}^T\boldsymbol{P}s_{v^*} + 2\boldsymbol{P}_{:,i}^T\boldsymbol{P}s_{v^*} + (\boldsymbol{P}^T\boldsymbol{P})_{i,i}. \tag{48}$$

In the above computations, we can precompute and store $\boldsymbol{P}s_k$, $s_k^T s_k$, $s_k^T\boldsymbol{P}^T\boldsymbol{P}s_k$ $(k = 1, 2, \ldots, c)$, and $(\boldsymbol{P}^T\boldsymbol{P})_{i,i}$ $(i = 1, 2, \ldots, m)$. Since $(\boldsymbol{P}^T\boldsymbol{P})_{i,i}$ does not need to be updated, during the c iterations, only $\boldsymbol{P}_{:,i}^T\boldsymbol{P}s_k$ needs to be recomputed c times from (43) to (48).

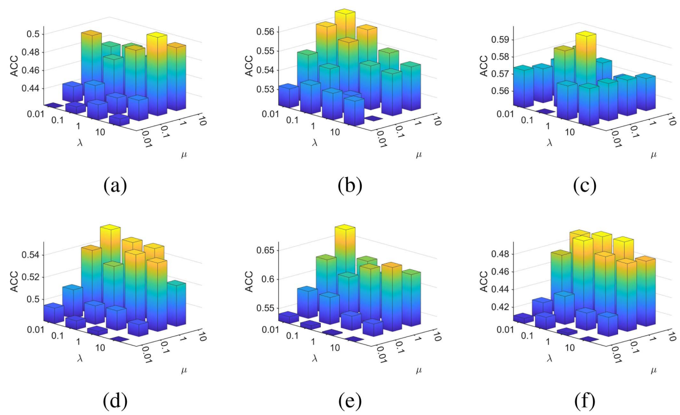


Fig. 5. The parameter sensitivity analysis in term of ACC. (a)–(f) Correspond to MiceProtein, ORL, Semeion, MSRA25, Olivetti, and Umist, respectively.

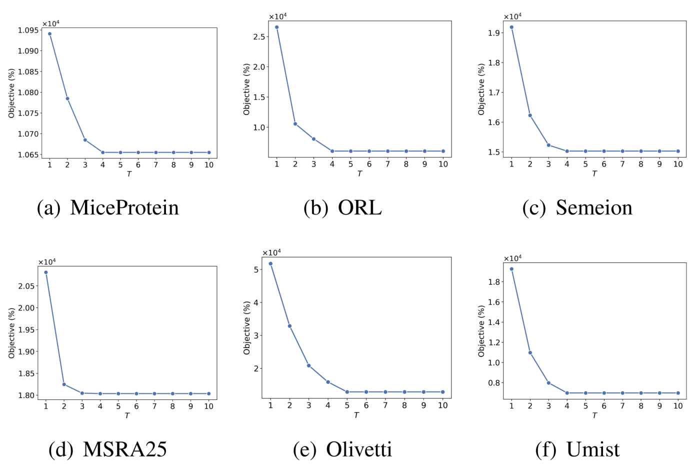


Fig. 6. Objective values monotonically decrease after each iteration.

Finally, the clustering indicator matrix for the original samples is derived using label propagation, expressed as $\boldsymbol{F} = \boldsymbol{MS}$. The detailed procedure for solving Problem (6) is outlined in Algorithm 1.

B. Computational Complexity

The LSPC-LA model operates through a preprocessing stage followed by Algorithm 1. In the preprocessing stage, the K-means++ algorithm is used to extract the local structure and construct the matrix $\boldsymbol{M}$, with a computational complexity of $\mathcal{O}(ndm)$. Algorithm 1 involves optimizing four variables, namely $\boldsymbol{P}$, $\boldsymbol{Q}$, $\boldsymbol{R}$ and $\boldsymbol{S}$. Here, we focus only on the dominant computational cost when optimizing each variable.

The main computational cost of optimizing $\boldsymbol{P}$ comes from computing $\boldsymbol{XQK}$, which incurs a complexity of $\mathcal{O}(ndm)$. The dominant computational cost in optimizing $\boldsymbol{Q}$ arises from computing $\{\boldsymbol{X}\boldsymbol{q}_i\}_{i=1}^m$ and $\boldsymbol{KP}^T\boldsymbol{X}$, each with a complexity of $\mathcal{O}(ndm)$. The computational cost of optimizing $\boldsymbol{R}$ and $\boldsymbol{S}$ is primarily determined by computing $\boldsymbol{PS}$, whose complexity is $\mathcal{O}(dmc)$. Since $d, m, c \ll n$, the overall computational complexity is $\mathcal{O}(ndm)$.

V. Experiments

In this section, we conduct a comprehensive set of experiments to demonstrate the superiority of LSPC-LA.

A. Experimental Settings

1) Dataset Descriptions: We used six real-world datasets to evaluate clustering performance. Detailed descriptions of the datasets are as follows:

- MSRA25,[2] Olivetti,[3] ORL[4] and Umist644[5] consist of different facial images.
- Mice Protein[6] comprises expression levels of 77 proteins measured in the cortex of control and Down syndrome mice exposed to context fear conditioning, spanning 8 classes.
- Semeion[7] includes 1,593 handwritten digit samples, featuring digits 0 to 9. Each digit was written twice on paper, first in a normal style and then in a faster style.

2) Evaluation Metrics and Comparison Models: To assess the quality and effectiveness of our clustering model, we employed five evaluation metrics: Purity, F1-score, Normalized Mutual Information (NMI), Accuracy (ACC), and Adjusted Rand Index (ARI). Each algorithm was run 30 times, and average values with standard deviations were recorded. We compared our proposed method with the following nine algorithms, which include both anchor-based and non-anchor-based clustering algorithms: KMNF [42], IRW-FCM [43], CPKM [44], CPKM2.0 [45], CAFCM [46], FCAG [28], EDCAG [47], FSSC [20], and FSEC [48].

3) Hyperparameter Settings: For each dataset, the number of anchors is controlled as a multiple of the number of dataset categories, denoted as T. We employ a grid search technique to find the optimal value of T for each dataset. Specifically, for the MSRA25 and Olivetti datasets, the grid search range is $\{10, 15, \ldots, 35\}$ with a step size of 5; for the ORL32x32 dataset, the range is $\{3, 4, \ldots, 8\}$ with a step size of 1; for the Umist644 dataset, the range is $\{3, 6, \ldots, 18\}$ with a step size of 3; for the Semeion dataset, the range is $\{25, 30, \ldots, 50\}$ with a step size of 5; and for the Mice Protein dataset, the range is $\{40, 45, \ldots, 65\}$ with a step size of 5.

We also use the grid search technique, setting λ to vary in the range of $\{10^{-2}, 10^{-1}, 10^{0}, 10^{1}\}$ and μ vary in the range of $\{10^{-2}, 10^{-1}, 10^{0}, 10^{1}\}$. To ensure a fair comparison, we performed the grid search using the provided code and parameter settings by the respective authors, and displayed the optimal results obtained. All experiments were meticulously performed on the MATLAB 2021b, equipped with 13th Gen Intel(R) Core(TM) Intel i7-13700 CPU and 32 GB of RAM.

B. Comparison Experiments

Table II summarizes the clustering performance of various algorithms across six real-world datasets. LSPC-LA demonstrates

[2]https://github.com/codefly13/fs_dataset/tree/master
[3]https://cam-orl.co. UK/facedatabase.html
[4]https://cam-orl.co. UK/facedatabase.html
[5]https://cs.nyu.edu/\;roweis/data/umist_cropped.mat
[6]https://doi.org/10.1371/journal.pone.0129126.s005
[7]https://archive.ics.uci.edu/dataset/178/semeion+handwritten+digit

TABLE II
CLUSTERING PERFORMANCE COMPARISON ON REAL-WORLD BENCHMARK DATASETS

Metric	Purity	F1-score	ACC	NMI	ARI	Purity	F1-score	ACC	NMI	ARI
	MSRA25					Olivetti				
KMNF	0.535$_{\pm0.031}$	0.401$_{\pm0.034}$	0.506$_{\pm0.038}$	0.557$_{\pm0.035}$	0.341$_{\pm0.037}$	0.444$_{\pm0.027}$	0.330$_{\pm0.021}$	0.417$_{\pm0.025}$	0.403$_{\pm0.019}$	0.254$_{\pm0.024}$
IRW-FCM	0.552$_{\pm0.026}$	0.413$_{\pm0.034}$	0.529$_{\pm0.027}$	0.559$_{\pm0.028}$	0.354$_{\pm0.040}$	0.467$_{\pm0.023}$	0.350$_{\pm0.019}$	0.443$_{\pm0.023}$	0.417$_{\pm0.020}$	0.277$_{\pm0.021}$
CPKM	0.468$_{\pm0.000}$	0.386$_{\pm0.000}$	0.446$_{\pm0.000}$	0.501$_{\pm0.000}$	0.319$_{\pm0.000}$	0.456$_{\pm0.026}$	0.350$_{\pm0.020}$	0.441$_{\pm0.026}$	0.418$_{\pm0.022}$	0.276$_{\pm0.022}$
CPKM2.0	0.576$_{\pm0.000}$	0.453$_{\pm0.000}$	0.540$_{\pm0.000}$	0.602$_{\pm0.000}$	0.400$_{\pm0.000}$	0.502$_{\pm0.000}$	0.386$_{\pm0.001}$	0.489$_{\pm0.000}$	0.456$_{\pm0.000}$	0.315$_{\pm0.001}$
CAFCM	0.575$_{\pm0.000}$	0.453$_{\pm0.000}$	0.540$_{\pm0.000}$	0.601$_{\pm0.000}$	0.399$_{\pm0.000}$	0.502$_{\pm0.000}$	0.385$_{\pm0.000}$	0.489$_{\pm0.000}$	0.456$_{\pm0.000}$	0.315$_{\pm0.000}$
FCAG	0.546$_{\pm0.033}$	0.443$_{\pm0.042}$	0.516$_{\pm0.041}$	0.631$_{\pm0.039}$	0.385$_{\pm0.050}$	0.544$_{\pm0.041}$	0.443$_{\pm0.043}$	0.516$_{\pm0.054}$	0.558$_{\pm0.029}$	0.380$_{\pm0.048}$
EDCAG	0.546$_{\pm0.017}$	0.433$_{\pm0.022}$	0.508$_{\pm0.013}$	0.619$_{\pm0.024}$	0.377$_{\pm0.024}$	0.549$_{\pm0.033}$	0.431$_{\pm0.029}$	0.518$_{\pm0.030}$	0.538$_{\pm0.029}$	0.366$_{\pm0.033}$
FSSC	0.522$_{\pm0.015}$	0.395$_{\pm0.014}$	0.503$_{\pm0.026}$	0.540$_{\pm0.017}$	0.334$_{\pm0.017}$	0.577$_{\pm0.025}$	0.442$_{\pm0.019}$	0.549$_{\pm0.026}$	0.526$_{\pm0.019}$	0.376$_{\pm0.022}$
FSEC	0.571$_{\pm0.017}$	0.449$_{\pm0.028}$	0.539$_{\pm0.035}$	0.638$_{\pm0.018}$	0.393$_{\pm0.033}$	0.532$_{\pm0.033}$	0.404$_{\pm0.031}$	0.512$_{\pm0.033}$	0.499$_{\pm0.036}$	0.333$_{\pm0.034}$
Ours	**0.597**$_{\pm0.021}$	**0.500**$_{\pm0.032}$	**0.552**$_{\pm0.025}$	**0.689**$_{\pm0.025}$	**0.449**$_{\pm0.037}$	**0.677**$_{\pm0.026}$	**0.583**$_{\pm0.033}$	**0.665**$_{\pm0.033}$	**0.679**$_{\pm0.031}$	**0.454**$_{\pm0.037}$
	Umist					Semeion				
KMNF	0.482$_{\pm0.026}$	0.334$_{\pm0.024}$	0.409$_{\pm0.026}$	0.601$_{\pm0.019}$	0.296$_{\pm0.026}$	0.468$_{\pm0.039}$	0.336$_{\pm0.025}$	0.433$_{\pm0.041}$	0.431$_{\pm0.021}$	0.252$_{\pm0.027}$
IRW-FCM	0.512$_{\pm0.015}$	0.352$_{\pm0.013}$	0.426$_{\pm0.012}$	0.619$_{\pm0.13}$	0.316$_{\pm0.014}$	0.490$_{\pm0.040}$	0.361$_{\pm0.034}$	0.464$_{\pm0.047}$	0.458$_{\pm0.025}$	0.284$_{\pm0.039}$
CPKM	0.494$_{\pm0.016}$	0.345$_{\pm0.012}$	0.413$_{\pm0.009}$	0.618$_{\pm0.007}$	0.309$_{\pm0.013}$	0.453$_{\pm0.019}$	0.328$_{\pm0.021}$	0.428$_{\pm0.013}$	0.437$_{\pm0.023}$	0.248$_{\pm0.026}$
CPKM2.0	0.504$_{\pm0.000}$	0.350$_{\pm0.000}$	0.421$_{\pm0.000}$	0.618$_{\pm0.000}$	0.313$_{\pm0.000}$	0.485$_{\pm0.011}$	0.330$_{\pm0.007}$	0.438$_{\pm0.014}$	0.434$_{\pm0.008}$	0.247$_{\pm0.010}$
CAFCM	0.504$_{\pm0.000}$	0.350$_{\pm0.000}$	0.421$_{\pm0.000}$	0.618$_{\pm0.000}$	0.313$_{\pm0.000}$	0.490$_{\pm0.001}$	0.329$_{\pm0.004}$	0.453$_{\pm0.020}$	0.434$_{\pm0.007}$	0.245$_{\pm0.005}$
FCAG	0.503$_{\pm0.015}$	0.373$_{\pm0.010}$	0.443$_{\pm0.017}$	0.640$_{\pm0.012}$	0.337$_{\pm0.012}$	0.509$_{\pm0.036}$	0.373$_{\pm0.036}$	0.481$_{\pm0.033}$	0.464$_{\pm0.043}$	0.297$_{\pm0.042}$
EDCAG	0.506$_{\pm0.016}$	0.359$_{\pm0.018}$	0.439$_{\pm0.009}$	0.623$_{\pm0.013}$	0.322$_{\pm0.020}$	0.476$_{\pm0.029}$	0.362$_{\pm0.040}$	0.461$_{\pm0.034}$	0.446$_{\pm0.040}$	0.287$_{\pm0.046}$
FSSC	0.494$_{\pm0.018}$	0.344$_{\pm0.008}$	0.424$_{\pm0.013}$	0.594$_{\pm0.011}$	0.306$_{\pm0.008}$	0.597$_{\pm0.050}$	0.468$_{\pm0.054}$	0.557$_{\pm0.062}$	0.552$_{\pm0.041}$	**0.404**$_{\pm0.062}$
FSEC	0.525$_{\pm0.023}$	0.380$_{\pm0.021}$	0.453$_{\pm0.014}$	0.634$_{\pm0.015}$	0.345$_{\pm0.022}$	0.579$_{\pm0.039}$	0.436$_{\pm0.035}$	0.538$_{\pm0.051}$	0.531$_{\pm0.027}$	0.367$_{\pm0.041}$
Ours	**0.555**$_{\pm0.018}$	**0.444**$_{\pm0.024}$	**0.496**$_{\pm0.026}$	**0.682**$_{\pm0.011}$	**0.411**$_{\pm0.025}$	**0.617**$_{\pm0.025}$	**0.469**$_{\pm0.018}$	**0.598**$_{\pm0.029}$	**0.558**$_{\pm0.018}$	**0.404**$_{\pm0.020}$
	Mice Protein					ORL32x32				
KMNF	0.448$_{\pm0.017}$	0.329$_{\pm0.010}$	0.422$_{\pm0.018}$	0.347$_{\pm0.016}$	0.219$_{\pm0.010}$	0.545$_{\pm0.029}$	0.347$_{\pm0.035}$	0.497$_{\pm0.033}$	0.696$_{\pm0.020}$	0.329$_{\pm0.037}$
IRW-FCM	0.466$_{\pm0.019}$	0.336$_{\pm0.022}$	0.424$_{\pm0.022}$	0.361$_{\pm0.025}$	0.233$_{\pm0.026}$	0.608$_{\pm0.022}$	0.428$_{\pm0.025}$	0.562$_{\pm0.027}$	0.748$_{\pm0.013}$	0.413$_{\pm0.026}$
CPKM	0.468$_{\pm0.001}$	0.335$_{\pm0.000}$	0.450$_{\pm0.001}$	0.356$_{\pm0.001}$	0.224$_{\pm0.000}$	0.612$_{\pm0.000}$	**0.457**$_{\pm0.000}$	0.569$_{\pm0.000}$	0.661$_{\pm0.000}$	0.429$_{\pm0.000}$
CPKM2.0	0.472$_{\pm0.000}$	0.338$_{\pm0.000}$	0.452$_{\pm0.000}$	0.363$_{\pm0.000}$	0.227$_{\pm0.000}$	0.615$_{\pm0.012}$	0.443$_{\pm0.011}$	0.568$_{\pm0.008}$	0.751$_{\pm0.010}$	0.429$_{\pm0.012}$
CAFCM	0.472$_{\pm0.000}$	0.338$_{\pm0.000}$	0.452$_{\pm0.000}$	0.363$_{\pm0.000}$	0.227$_{\pm0.000}$	0.618$_{\pm0.000}$	0.443$_{\pm0.000}$	0.570$_{\pm0.000}$	0.751$_{\pm0.000}$	0.429$_{\pm0.000}$
FCAG	0.526$_{\pm0.026}$	0.383$_{\pm0.014}$	0.495$_{\pm0.033}$	0.435$_{\pm0.016}$	0.283$_{\pm0.015}$	0.597$_{\pm0.032}$	0.441$_{\pm0.038}$	0.559$_{\pm0.037}$	0.752$_{\pm0.021}$	0.427$_{\pm0.039}$
EDCAG	0.502$_{\pm0.023}$	0.373$_{\pm0.031}$	0.466$_{\pm0.033}$	0.425$_{\pm0.041}$	0.278$_{\pm0.036}$	0.563$_{\pm0.022}$	0.379$_{\pm0.020}$	0.527$_{\pm0.022}$	0.724$_{\pm0.012}$	0.363$_{\pm0.021}$
FSSC	**0.541**$_{\pm0.053}$	**0.431**$_{\pm0.041}$	0.507$_{\pm0.57}$	**0.534**$_{\pm0.033}$	**0.361**$_{\pm0.046}$	0.588$_{\pm0.021}$	0.411$_{\pm0.026}$	0.552$_{\pm0.018}$	0.732$_{\pm0.015}$	0.395$_{\pm0.027}$
FSEC	0.521$_{\pm0.021}$	0.354$_{\pm0.023}$	0.442$_{\pm0.022}$	0.421$_{\pm0.020}$	0.266$_{\pm0.026}$	0.461$_{\pm0.020}$	0.306$_{\pm0.022}$	0.434$_{\pm0.018}$	0.648$_{\pm0.016}$	0.287$_{\pm0.022}$
Ours	0.538$_{\pm0.027}$	0.423$_{\pm0.035}$	**0.511**$_{\pm0.021}$	0.497$_{\pm0.035}$	0.342$_{\pm0.042}$	**0.619**$_{\pm0.017}$	0.453$_{\pm0.030}$	**0.572**$_{\pm0.019}$	**0.758**$_{\pm0.031}$	**0.439**$_{\pm0.014}$

†:Best results are in bold, and the second best results are underlined.

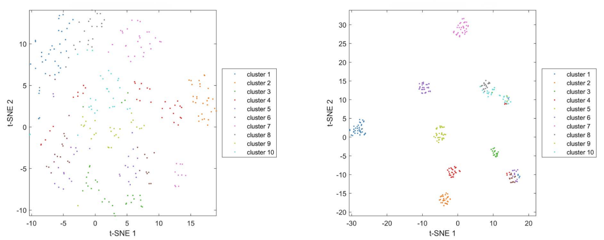
the best or second-best performance across five metrics on all six datasets. It significantly outperforms other algorithms on the MSRA25, Olivetti, Semeion, and Umist datasets, showcasing remarkable advantages. Specifically, on the Olivetti and Umist datasets, LSPC-LA achieves improvements of 11.6% and 4.3% in ACC, 14.1% and 6.4% in NMI, and 7.4% and 6.6% in ARI, respectively, compared to the second-best algorithm. These results highlight the applicability of LSPC-LA on these datasets. Overall, the performance on multiple real-world datasets demonstrates LSPC-LA's excellent clustering ability, with superior generalization and stability.

C. Ablation Experiments

We conducted an ablation study, and the results are presented in Table III. The first scenario removes the prior indicator matrix $\boldsymbol{M}$, eliminating prior knowledge and preventing the learning of the anchor guiding matrix $\boldsymbol{Q}$. As a result, the local structure information is lost, and the algorithm degenerates into K-means. The second scenario substitutes the anchor guiding matrix $\boldsymbol{Q}$ with the matrix $\boldsymbol{M}$ without allowing $\boldsymbol{Q}$ to learn. In the third scenario, $\boldsymbol{Q}$ is allowed to learn but without regularization. The last scenario is the complete LSPC-LA model. The results show that narrowing the solution space through local structure effectively enhances clustering performance. When the anchor guiding matrix is learnable, it significantly improves the accuracy of anchor-based clustering. This highlights the key role of anchors in learning the clustering structure of original samples to improve clustering performance. Further regularization of $\boldsymbol{Q}$ leads to additional performance improvements.

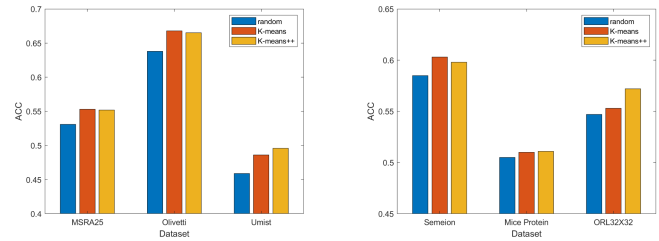
TABLE III
ABLATION EXPERIMENT

M	Q	Reg	ACC	Purity	NMI	ARI	ACC	Purity	NMI	ARI
			MSRA25				**Olivetti**			
×	-	-	0.507	0.531	0.494	0.273	0.457	0.481	0.522	0.365
√	×	-	0.529	0.548	0.581	0.311	0.468	0.595	0.578	0.391
√	√	×	0.533	0.562	0.658	0.386	0.615	0.635	0.625	0.415
√	√	√	0.552	0.597	0.689	0.449	0.665	0.677	0.679	0.454
			Umist				**Semeion**			
×	×	-	0.428	0.445	0.419	0.251	0.456	0.475	0.462	0.266
√	×	-	0.433	0.450	0.453	0.272	0.486	0.505	0.488	0.281
√	√	×	0.479	0.500	0.621	0.399	0.566	0.585	0.491	0.390
√	√	√	0.496	0.555	0.682	0.411	0.598	0.617	0.558	0.404
			Mice Protein				**ORL32x32**			
×	×	-	0.439	0.455	0.213	0.243	0.538	0.555	0.661	0.344
√	×	-	0.443	0.460	0.246	0.261	0.563	0.585	0.690	0.350
√	√	×	0.484	0.505	0.313	0.322	0.548	0.565	0.720	0.412
√	√	√	0.511	0.538	0.497	0.342	0.572	0.619	0.758	0.439



(a) Before iteration (b) After iteration (convergence)

Fig. 7. T-SNE visualization experiment on Olivetti dataset.



(a) (b)

Fig. 8. Comparative experiment on anchor construction methods.

D. Local Structure Information Fineness

We assessed the accuracy of the local structure information and its impact on clustering performance with respect to the number of anchors. The number of anchors was set as T times the number of sample categories, and T was adjusted accordingly. The accuracy, termed "Fineness", is defined as $Fineness = \frac{CPTP}{PTP}$, where PTP represents the total number of sample pairs, and $CPTP$ is the number of correct sample pairs, which are Must-Link pairs belonging to the same class in the original labels. The results in Fig. 4 show that for all datasets, the fineness of the local structure information exceeds 90% once the number of anchors surpasses a certain threshold, confirming the accuracy of the local structure information.

E. Impact for Parameters

Both λ and μ are hyperparameters that balance the importance of different modules in LSPC-LA, varying within the range $[10^{-2}, 10^{1}]$. As shown in Fig. 5, the clustering accuracy bar charts reveal that when $\mu \in \{10^0, 10^1\}$ and $\lambda \in \{10^{-2}, 10^{-1}\}$, our model achieves excellent clustering performance across all datasets, demonstrating the stability of the model.

F. Convergence Analysis

As shown in Fig. 6, we present the convergence curves of the objective function of the proposed LSPC-LA model across all experimental datasets. It can be clearly observed that, for each dataset, the value of the objective function exhibits a significant monotonic decreasing trend, particularly with a rapid decline in the early stages of training. As the iterations proceed, the objective value gradually stabilizes around a low level without noticeable fluctuations, indicating that the model has reached a convergent state. This phenomenon, visualized through empirical results, consistently and convincingly demonstrates the favorable convergence properties of the proposed LSPC-LA optimization algorithm.

G. Visualization

We visualize the anchor distribution on the Olivetti dataset using t-SNE (Fig. 7). Each point is an anchor, colored by the majority ground-truth class of the samples it represents (ties broken by the smallest class index). Before optimization, anchors show no clear clustering structure. After optimization (at convergence), anchors form well-separated groups that align with the underlying classes. Moreover, intra-class distances shrink and inter-class distances increase, indicating improved geometric alignment with cluster structure.

H. Impact of the Anchor Construction Method

We tested anchor construction methods using different strategies and compared their impact on the final clustering results, as shown in Fig. 8. The results indicate that, compared with random selection, both K-means++ and K-means better capture the underlying sample structure. This allows the model to more effectively exploit local structural information to guide the clustering process, ultimately yielding superior clustering performance.

VI. Conclusion and Further Discussion

A. Conclusion

This paper enhances K-means by leveraging local structural information. Anchors extract Must-Link information, narrowing the solution space and reducing the likelihood of converging to poor local minima. By learning sample structures, anchors adaptively adjust to maintain clustering consistency, improving performance. Compared to existing methods, LSPC-LA better utilizes structural information, achieving superior results. Extensive experiments validate the accuracy of local structure information and the superior clustering performance of LSPC-LA.

B. Further Discussion

The performance of LSPC-LA is constrained by its reliance on the local linearity assumption in Euclidean space, which may hinder its ability to accurately capture local structural information when dealing with data exhibiting complex manifold structures or strong nonlinear characteristics. In addition, the final clustering performance is partially influenced by the quality of the initial anchor set. Although the learnable anchor mechanism can substantially correct suboptimal initial anchors, errors may still propagate in challenging scenarios, such as when the initial anchors deviate significantly from true cluster centers or when substantial inter-class overlap exists. These limitations highlight directions for future research, as investigating them can help enhance the scalability and extensibility of our model.

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